TD1 – Structures arborescentes

Exercice 2

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Q1 / Un polynôme est un tableau de monômes :

$$x^4 + 3x^{100} \rightsquigarrow \begin{array}{|c|c|} \hline 0 & 1 \\ \hline x^4 & 3x^{100} \\ \hline \end{array}$$

```
def polynome(p: Array[monomes]) -> None:
    int nb_termes = len(p)

    for i in range(0, nb_termes):

        p[i].coeff = p[i].degre * p[i].coeff
        if p[i].degre != 0:
            p[i].degre = p[i].degre - 1
```

Q2 / $x \mapsto x^4 + 3x^{100}$

Représentation 1

$$\text{monomes} \rightsquigarrow \begin{array}{|c|c|} \hline 0 & 1 \\ \hline \begin{cases} \text{coeff} & 3 \\ \text{degré} & 100 \end{cases} & \begin{cases} \text{coeff} & 1 \\ \text{degré} & 4 \end{cases} \\ \hline \end{array}$$

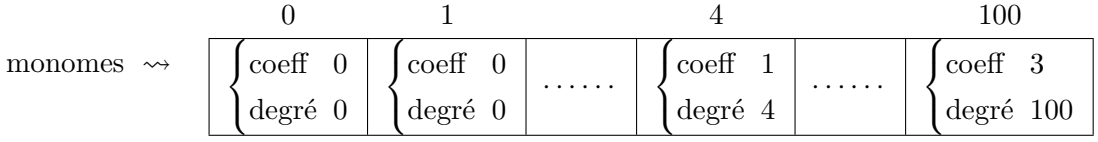
Représentation 2 (1 sentinelle)

$$\text{monomes} \rightsquigarrow \begin{array}{|c|c|c|} \hline 0 & 1 & 2 \\ \hline \begin{cases} \text{coeff} & 0 \\ \text{estSent} & 1 \\ \text{degré} & 0 \end{cases} & \begin{cases} \text{coeff} & 3 \\ \text{estSent} & 0 \\ \text{degré} & 100 \end{cases} & \begin{cases} \text{coeff} & 1 \\ \text{estSent} & 0 \\ \text{degré} & 4 \end{cases} \\ \hline \end{array}$$

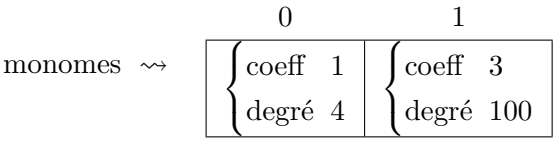
Représentation 3 (2 sentinelles)

$$\text{monomes} \rightsquigarrow \begin{array}{|c|c|c|c|} \hline 0 & 1 & 2 & 3 \\ \hline \begin{cases} \text{coeff} & 0 \\ \text{estSent} & 1 \\ \text{degré} & 0 \end{cases} & \begin{cases} \text{coeff} & 3 \\ \text{estSent} & 0 \\ \text{degré} & 100 \end{cases} & \begin{cases} \text{coeff} & 1 \\ \text{estSent} & 0 \\ \text{degré} & 4 \end{cases} & \begin{cases} \text{coeff} & 0 \\ \text{estSent} & 1 \\ \text{degré} & 0 \end{cases} \\ \hline \end{array}$$

Représentation 4



Q3 / Représentation canonique du polynôme $x \mapsto 1 \cdot x^{100} + 0 \cdot x^6 + x^4 + 2 \cdot x^{100}$:



Q4 /

Auxiliaire 1 : vérifier le coeff non nul.

Auxiliaire 2 : vérifier si le degré est déjà dans la table et additionner leurs coefficients.

Auxiliaire 3 : tri des monômes par degré croissant.

Q5 /

```
def sont_egaux(pol1: Array[monomes], pol2: Array[monomes]) -> bool:

    pol1 = canonique(pol1)
    pol2 = canonique(pol2)

    int t1 = size(pol1)
    int t2 = size(pol2)
    if t1 != t2:
        return False
    int res = True
    for i in range(0, t1):

        if (pol1[i].degre != pol2[i].degre) or (pol1[i].coeff != pol2[i].coeff):
            res = False

    return res
```